AGENTIA™ HEADLESS VIRTUAL HACKATHON · 2026

Agentia Failure Doctor

From a red Copado promotion to root cause, a confirmed fix, and the same failure never reaching the pipeline again. Headless: terminal, CI, Slack and AI agents.

- oclif plugin for the Agentia CLI
- MCP server · 6 tools
- Agent skill
- 40 tests

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THE PROBLEM

When a promotion fails, the fix is rarely hard. Finding it is.

Hunt

Open the job execution, click through steps, download result files, read raw Salesforce deploy errors.

Cross-check

Is the field in another user story? A merge conflict? An expired credential? Git, story and destination org, checked by hand.

Repeat

The same few patterns come back every week, investigated from scratch. The knowledge sits with two or three senior people.

Release trains stall, back-promotions pile up, and nobody can quickly say what keeps failing or how long recovery takes.

THE SOLUTION

One loop, five headless commands

1 · Diagnose

`doctor diagnose`

root cause vs symptoms, evidence, confidence

2 · Fix

`doctor fix --apply`

exact Agentia commands, confirmed one by one

3 · Prevent

`doctor preflight`

blocks the failure before promotion

4 · Learn

`doctor learn`

a confirmed fix becomes a team rule

5 · Measure

`doctor dora · digest`

CFR, MTTR, deploy frequency, Slack

Built only on public `agentia … --json` commands: jobs, logs, conflicts, metadata dependencies, work items, promotions and Copado AI agents. Every command also runs with `--demo` against a synthetic dev → UAT → prod pipeline.

scan What is failing, grouped by root cause, not by job

zsh — acme-salesforce

```
$ agentia doctor scan --demo
20 failed job(s), 6 distinct root cause(s)

  8 × Apex references a class, method or variable that is missing  [missing-dependency]
      SF-MISSING-APEX-SYMBOL:Tax_Rate__c
      JE-000101, JE-001068, JE-001059, JE-001050, JE-001038, JE-001029, …
  8 × Transient platform error  [transient]
      SF-TRANSIENT:Invoice__c
      JE-001076, JE-001064, JE-001055, JE-001046, JE-001034, JE-001025, …
  1 × Unrecognised error  [unknown]
      UNKNOWN:Console apps require the Service Cloud User feature license on the deploying user
      JE-000106
  1 × Apex test assertion failed  [test-failure]
      SF-TEST-ASSERT:DiscountServiceTest
      JE-000104
  1 × Environment credential expired or invalid  [credential]
      ENV-CREDENTIAL:*
      JE-000103
  1 × Git merge conflict on the promotion branch  [merge-conflict]
      GIT-MERGE-CONFLICT:Account
      JE-000102
```

diagnose Root cause first; the cascade is labelled as symptoms

zsh — acme-salesforce

```
$ agentia doctor diagnose JE-000101 --demo
Job JE-000101 (a4N000000000101AAA) · Error · Promotion Deploy
Failed step: Deploy

ROOT CAUSE  Apex references a class, method or variable that is missing   [SF-MISSING-APEX-SYMBOL, confidence: high]
            missing/offending: CustomField Tax_Rate__c
            failing component(s): ApexClass:InvoiceService
            Apex compiled against something the destination does not have. Either a class in another story, or a field/variable change that did not
travel with this story.
            log: Error  ApexClass  InvoiceService  Variable does not exist: Tax_Rate__c (42:9)

ALSO SEEN (2), usually symptoms of the root cause:
  - Code coverage below the required 75% ×1
  - Dependent class is invalid (cascade) (ApexClass:InvoiceBatch, ApexClass:InvoiceController) ×2

EVIDENCE
  [dependencies] ApexClass InvoiceService depends on CustomField:Invoice__c.Tax_Rate__c, which is missing from the destination

NEXT STEPS  (🔍 read-only · ✏️  changes Copado, needs confirmation · 👤 you)
  1. 👤 Find where CustomField Tax_Rate__c lives
     The destination is missing something this story depends on. Locate the story that committed it (e.g. `agentia cicd metadata list --filter-type
CustomField --source PreviouslyCommitted --pipeline-id <id> --source-org-id <id>`), or confirm nobody committed it.
  2. 👤 Promote the owning story first, or add the component to this story
     Either order the stories so the dependency lands first, or commit it alongside this change (then re-promote).
```

fix A merge conflict: dry-run plan, every change confirmed

zsh — acme-salesforce

```
$ agentia doctor fix JE-000102 --apply --demo
Job JE-000102 (a4N000000000102AAA) · Error · Promotion Deploy
Failed step: Merge

ROOT CAUSE  Git merge conflict on the promotion branch   [GIT-MERGE-CONFLICT, confidence: high]
            failing component(s): Layout:Account
            Two stories changed the same file. Copado stops the promotion until each conflict is resolved, then the same execution can resume.
            log: CONFLICT (content): Merge conflict in force-app/main/default/layouts/Account-Account Layout.layout-meta.xml

ALSO SEEN (1), usually symptoms of the root cause:
  - Git merge conflict on the promotion branch (PermissionSet:Sales_Ops) ×1

EVIDENCE
  [conflicts] 2 unresolved of 2 conflict(s) on promotion a0B000000000102AAA

NEXT STEPS  (🔍 read-only · ✏️  changes Copado, needs confirmation · 👤 you)
  1. 🔍 Review conflict a0CDEMO0000000001AA
     Read the conflicting content before choosing a resolution.
     $ agentia cicd promotion conflict get a0CDEMO0000000001AA -p a0B000000000102AAA
  2. ✏️  Auto-resolve conflict a0CDEMO0000000001AA
     Let Copado apply its automatic resolution. Use --mode manual --file <path> instead for reviewed content.
     $ agentia cicd promotion conflict resolve a0CDEMO0000000001AA -p a0B000000000102AAA --mode auto
  3. 🔍 Review conflict a0CDEMO0000000002AA
     Read the conflicting content before choosing a resolution.
     $ agentia cicd promotion conflict get a0CDEMO0000000002AA -p a0B000000000102AAA
  4. ✏️  Auto-resolve conflict a0CDEMO0000000002AA
     Let Copado apply its automatic resolution. Use --mode manual --file <path> instead for reviewed content.
     $ agentia cicd promotion conflict resolve a0CDEMO0000000002AA -p a0B000000000102AAA --mode auto
  5. ✏️  Resume the same promotion execution
     After conflicts are resolved, resume rather than start a new run.
     $ agentia cicd promotion run a0B000000000102AAA --resume a4N000000000102AAA
```

`fix --apply` Resolve both conflicts, resume the same execution, verify

zsh — acme-salesforce

```
$ agentia doctor fix JE-000102 --apply --demo
…
✔ Review conflict a0CDEMO0000000001AA

Auto-resolve conflict a0CDEMO0000000001AA
  $ agentia cicd promotion conflict resolve a0CDEMO0000000001AA -p a0B000000000102AAA --mode auto
Run this? [y/N] y
✔ Auto-resolve conflict a0CDEMO0000000001AA
✔ Review conflict a0CDEMO0000000002AA

Auto-resolve conflict a0CDEMO0000000002AA
  $ agentia cicd promotion conflict resolve a0CDEMO0000000002AA -p a0B000000000102AAA --mode auto
Run this? [y/N] y
✔ Auto-resolve conflict a0CDEMO0000000002AA

Resume the same promotion execution
  $ agentia cicd promotion run a0B000000000102AAA --resume a4N000000000102AAA
Run this? [y/N] y
✔ Resume the same promotion execution

$ agentia doctor diagnose JE-000102 --demo
Job JE-000102 (a4N000000000102AAA) · Success · Promotion Deploy

✔ Job succeeded. Nothing to fix.
```

`--ai` Unknown error? Rules first, Copado AI agent second

zsh — acme-salesforce

```
$ agentia doctor diagnose JE-000106 --ai --demo
Job JE-000106 (a4N000000000106AAA) · Error · Promotion Deploy
Failed step: Deploy

ROOT CAUSE  Unrecognised error   [UNKNOWN, confidence: medium]
            failing component(s): CustomApplication:Acme_Service_Console
            No rule matched this error. Run with --ai to ask a Copado AI agent, then record the answer with `doctor learn`.
            log: Error  CustomApplication  Acme_Service_Console  Console apps require the Service Cloud User feature license on the deploying user (0:0)

EVIDENCE
  [ai] Root cause: the deploying user has no Service Cloud User feature licence, so Salesforce rejects console apps (CustomApplication with a console
navigation). Fix: tick "Service Cloud User" on the integration user the UAT credential logs in as (Setup → Users), then resume the job. No metadata change is
needed.

NEXT STEPS  (🔍 read-only · ✏️  changes Copado, needs confirmation · 👤 you)
  1. 👤 Unrecognised error
     No rule matched this error. Run with --ai to ask a Copado AI agent, then record the answer with `doctor learn`.
```

learn The confirmed answer becomes a team rule, versioned in Git

zsh — acme-salesforce

```
$ agentia doctor learn JE-000106 --title "Console app needs Service Cloud licence" --from-ai --category access --demo
✔ Learned LOCAL-001: Console app needs Service Cloud licence
  match: /Console apps require the Service Cloud User feature license on the deploying user/i
  saved: .agentia-doctor/signatures.json
Commit it so the whole team gets the rule.

$ agentia doctor diagnose JE-000106 --demo
Job JE-000106 (a4N000000000106AAA) · Error · Promotion Deploy
Failed step: Deploy

ROOT CAUSE  Console app needs Service Cloud licence   [LOCAL-001, confidence: high]
            failing component(s): CustomApplication:Acme_Service_Console
            Root cause: the deploying user has no Service Cloud User feature licence, so Salesforce rejects console apps (CustomApplication with a console navigation).
Fix: tick "Service Cloud User" on the integration user the UAT credential logs in as (Setup → Users), then resume the job. No metadata change is needed.
            log: Error  CustomApplication  Acme_Service_Console  Console apps require the Service Cloud User feature license on the deploying user (0:0)
```

preflight

Stop the failure before it reaches the pipeline

zsh — acme-salesforce

```
$ agentia doctor preflight US-0000105 --demo
Preflight US-0000105: 3 component(s) checked against the destination
  ✖ BLOCK  Flow:Discount_Approval needs CustomField:Opportunity.Discount_Pct__c, which is not in the destination and not in US-0000105
           fix: Add CustomField:Opportunity.Discount_Pct__c to US-0000105, or promote the story that owns it first
  ✖ BLOCK  ApexClass:DiscountService needs CustomMetadata:Discount_Tier__mdt.Gold, which is not in the destination and not in US-0000105
           fix: Add CustomMetadata:Discount_Tier__mdt.Gold to US-0000105, or promote the story that owns it first
✖ 2 blocker(s). This promotion would fail; fix them first.

$ agentia doctor preflight US-0000107 --demo
Preflight US-0000107: 2 component(s) checked against the destination
✔ Safe to promote.

$ agentia doctor preflight --install-hook
✔ .git/hooks/pre-push
Set DOCTOR_PIPELINE_ID, DOCTOR_SOURCE_ORG_ID and DOCTOR_TARGET_ORG_ID for it.
```

`dora · digest`

Measure whether release health is improving

zsh — acme-salesforce

```
$ agentia doctor dora --demo
Window: 2026-09-14 → 2026-10-26 (42.25 days, 57 deployment jobs)
Deployment frequency   1.02 successful / day
Change failure rate    25%
Mean time to recovery  5.8 h  (10 recovered, 4 still failing)

By job type:
  Promotion Deploy                  14 failed / 57  (25%)
```

Failure Doctor APP 9:00 AM

Copado release health

Change failure rate
25%

Mean time to recovery
5.8 h

Deployments / day
1.02

Still failing
4 lane(s)

57 deployment jobs, 2026-09-14 → 2026-10-26

Top root causes (19 failed jobs, 5 distinct)

- **8×** Apex references a class, method or variable that is missing `missing-dependency`
  JE-000101, JE-001068, JE-001059, JE-001050, …
- **8×** Transient platform error `transient`
  JE-001076, JE-001064, JE-001055, JE-001046, …
- **1×** Console app needs Service Cloud licence `access`
  JE-000106
- **1×** Apex test assertion failed `test-failure`
  JE-000104
- **1×** Environment credential expired or invalid `credential`
  JE-000103

MCP Same tools for Claude, Cursor and Copado AI agents

zsh — acme-salesforce

```
$ agentia doctor mcp      # stdio MCP server, as Claude / Cursor / Copado AI would start it

→ tools/list
   doctor_diagnose      Diagnose a failed Copado job  readOnly=True
   doctor_plan_fix      Plan the fix for a failed Copado job  readOnly=True
   doctor_scan          Group recent Copado failures by root cause  readOnly=True
   doctor_dora          DORA metrics from Copado job history  readOnly=True
   doctor_signatures    List known failure signatures  readOnly=True
   doctor_preflight     Preflight a user story before promotion  readOnly=True

→ tools/call doctor_diagnose {"jobId": "JE-000103"}
   Job JE-000103 (a4N000000000103AAA) · Error · Promotion Deploy
   Failed step: Deploy

   ROOT CAUSE  Environment credential expired or invalid   [ENV-CREDENTIAL, confidence: high]
               Copado could not log into the environment. Nothing in the story is wrong: re-authenticate the credential, then resume the job.
               log: Error  Org  UAT  INVALID_SESSION_ID: Session expired or invalid (refresh token revoked for credential "UAT Integration")

   NEXT STEPS  (🔍 read-only · ✏️  changes Copado, needs confirmation · 👤 you)
     1. 🔍 Find the failing environment credential
        The job could not log into the org; the change itself is fine.
        $ agentia cicd credential list
     2. 👤 Re-authenticate it in the browser
        Run `agentia cicd environment auth web login <environment-id>` for the environment above (browser-only).
     3. ✏️  Resume the job
        Outstanding steps pick up from where they stopped.
        $ agentia cicd job resume a4N000000000103AAA
```

ARCHITECTURE & GUARDRAILS

Extends Agentia the way it is meant to be extended

Surfaces

Terminal · CI · Git pre-push · Slack · Claude / Cursor (MCP) · Agent skill

▼

agentia doctor · oclif plugin, installed with `agentia plugins install`

extract → classify (18 ranked signatures + learned) → evidence → plan → confirm

▼ public `agentia … --json` only

Copado CICD gateway + AI agents

job get/list/log · promotion conflict list/resolve/run --resume · metadata dependency list · work status · ai agent ask

Safe by design

Read-only by default. Every write is shown as the exact Agentia command and confirmed. Manual steps are never automated.

Deterministic first

Rules before AI: repeatable, offline, free. AI only for unknowns, and its answers can become rules.

Secrets never leave

Session Ids, tokens, auth URLs and webhooks are redacted from reports, JSON, MCP and Slack.

WHY IT MATTERS

A product view: what a Copado customer gets

Release managers

Minutes, not an afternoon, from red promotion to cause. One fix clears a whole group of failures.

Developers

Preflight catches the missing dependency at `git push`, before the promotion even exists.

Teams & leads

Tribal knowledge becomes versioned rules. New people on release duty get the same answer as the experts.

Copado

Proof that Agentia is a platform: plugin + MCP + skill, on top of pipelines, Quality Gates, AI agents and DORA.

Next: content compare in preflight · packaged as a Local Quality Gate · learned rules shared across a pipeline · PR comments · live-org validation of the gateway field mapping

TRY IT

Open source, runs in 30 seconds

```
git clone https://github.com/divaaker/agentia-failure-doctor
cd agentia-failure-doctor && npm install && npm run build
agentia plugins link .
agentia doctor demo --reset
agentia doctor scan --demo
```

Built

12 commands · 6 MCP tools · agent skill · Slack digest · 40 tests

Honest status

Shown on synthetic data. Exercised through the real Agentia CLI; live-org runs await a Copado API key.

Data

Every Id, story and log line in the repo and this video is synthetic.